

Microservice: IELTS Macquarie [deleted]

While others were dreaming about it - I was getting it done.



Document revision history:

Date	Revision	Status	Author	Summary of Changes
28-04-2022	1.1	Published	@Selvakumar Balraj	Solution for Macquarie integration

Service objective:

The objective of this diagram to auto login the mobile logged-in user into Macquarie in-app browser which is part of Mobile app. User data like email, first-name and last-name will be

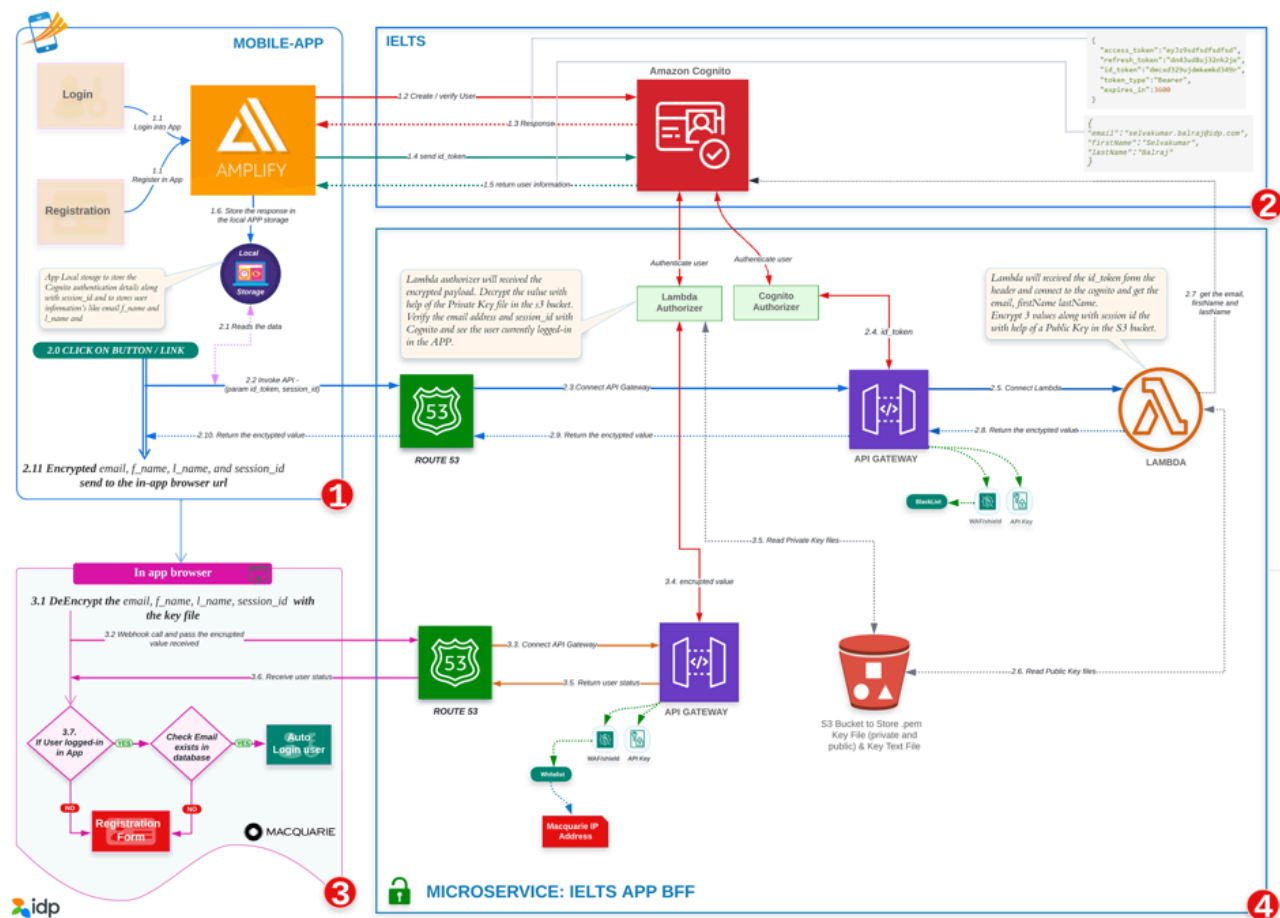
encrypted and gets shared to Macquarie via in-app browser.

About in-app browser:

In App Browser, in other words, the internal browser is an application provided by the operating system to which developers have access when programming their applications.

The In App Browser window behaves like a standard web browser, and can't access Cordova APIs. For this reason, the InAppBrowser is recommended if you need to load third-party (untrusted) content, instead of loading that into the main Cordova webview. The InAppBrowser is not subject to the whitelist, nor is opening links in the system browser. Its Supported Platforms like Android, Browser, iOS and Windows

Architecture diagram:



How this works?

#1. Login / register in Mobile:

1. If the user **login** into APP, verify the credentials with Cognito, if the credentials matched with Cognito, then will login the user into APP
2. If the user **Register** into APP, verify the email address in the Cognito, if the email address not exists, then will create the user into Cognito and auto login the user into the APP. if the email address exists in Cognito then raise error.
3. After login / Register in APP:
 - a. The below response data received from the Cognito will gets stored in the APP Local Storage.

```
1 {  
2   "access_token": "eyJz9sdfsd fsd",  
3   "refresh_token": "dn43ud8uj32nk2je",  
4   "id_token": "dmcxd329ujdmkemkd349r",  
5   "token_type": "Bearer",  
6   "expires_in": 3600  
7 }
```

#2. Once the #1 is completed, the user will start navigating to the pages within app, there will be a button/link in a page which will navigate to Macquarie website via in-app browser. **[URL is specific for IELTS By IDP Integration]**

1. When the user click on the link/button, will prompt for the confirmation message to get agreement from the user about sharing the data to the Macquarie, if the user agree to the terms, then the app will do a REST API call to IELTS APP Microservice and encrypt to encrypt below values. The encrypted value will gets passed to Macquarie via in-app browsers.

Payload template that need to pass to Macquarie in the encrypted format

```
1 {  
2   "emailAddress": "selvakumar.balraj@idp.com",  
3   "firstName": "Selvakumar",  
4   "lastName": "Balraj",  
5   "expiry_time": "dsPUg679KSSAEraYYrwn5aduu7iviSq899Rm1vN40fgpDs9F4loEUxsZt/ep//EU9+"  
6 }  
7  
8 Encrypted format:  
9 "cipherText":  
  "dsPUg679KSSAEraYYrwn5aduu7iviSq899Rm1vN40fgpDs9F4loEUxsZt/ep//EU9+iL5kAaiuFp5ZgeDThWQesM1b14  
  z7xvpy0lhtXM1t/i8ReL5t8WIfkbqAj7PxJZdycLZ9oVis4w0qJ8tRTc72Yy5u2H6FmNV8gKi00ncNq+XjpACxLEPz144  
  KQ8XMVmafMX25Vu10DP/1VYD1nCIQ9Mwwh5GwNNeTos0K1vBLQoHpp2Qg0E5G6Ha1AAKWxwacq2HsVTfUC1FZshm2+ks  
  Wt0A6NR0BC5MU21K9w8W2XdCIZ/R/YzGo8BhNm9cRCmTPMq320jeBgRP9G2gzAew==",
```

Encryption Process



IELTS APP will do a REST API Call, and pass below parameter values

→ id_token

→ Header

i Cognito authorizer configured in the API Gateway will get triggered

→ Validate the id_token

if the id_token not valid then

x Raise the error back to the APP - with the status **Authentication Failed**

i if the id_token is valid then do the below process

✓ Connects to the Lambda, Lambda will do the below action

→ Read the data "id_token" from the header, decrypt the id_token and get the uuid of the user

→ Pass the uuid to the Cognito and get the below details.

→ "emailaddress", "firstname" and "lastname"

→ Get the current date and time (dd/mm/yyyy hh:mm)

→ Reads the public .pem file from the S3 bucket (Separate key that needs to encrypt the date-time)

→ Encrypt the date time stamp

→ Read the Public .pem file from the S3 bucket (Separate Key that needs to encrypt the payload)

→ Encrypt the below Payload, with the public .pem key file.

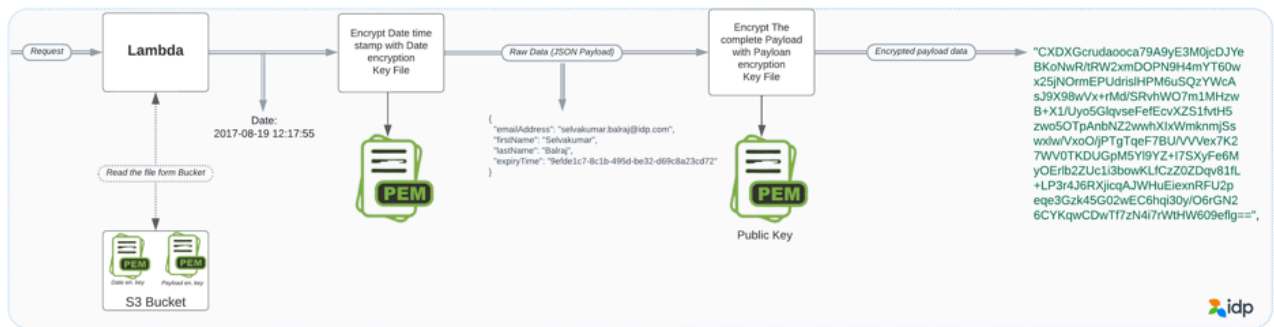
```
{  
  "email"="selvakumar.balraj@idp.com"  
  "firstName"="Selvakumar"  
  "lastName"="Balraj"  
  "expiryTime"="sdfsd29u3udemmkd34ru3udem"  
}
```

→ Return back the encrypted data to the APP

Example encrypted payload :

```
"CXDXGcrudaoooca79A9yE3M0jcDJYeBKoNwR/tRW2xmDOPN9H4mYT60w  
x25jN0rmEPUdris1HPM6uSQzYWcAsJ9X98wVx+rMd/SRvhW07m1MHzwB  
+X1/Uyo5G1qvseFefEcvXZS1fvth5zwo50TpAnbNZ2wwhXIxWmknmjSs"
```

wx1w/Vxo0/jPTgTqeF7BU/VVVex7K27WV0TKDUGpM5Y19YZ+I7SXYFe6
My0Er1b2ZUc1i3bowKLfCzZ0ZDqv81fL+LP3r4J6RXjicqAJWHuEiexn
RFU2peqe3Gzk45G02wEC6hqi30y/06rGN26CYKqwCDwTf7zN4i7rWtHW
609eflg==" ,



Once the encrypted payload is ready, the APP will pass the encrypted payload to Macquarie via in-app browser

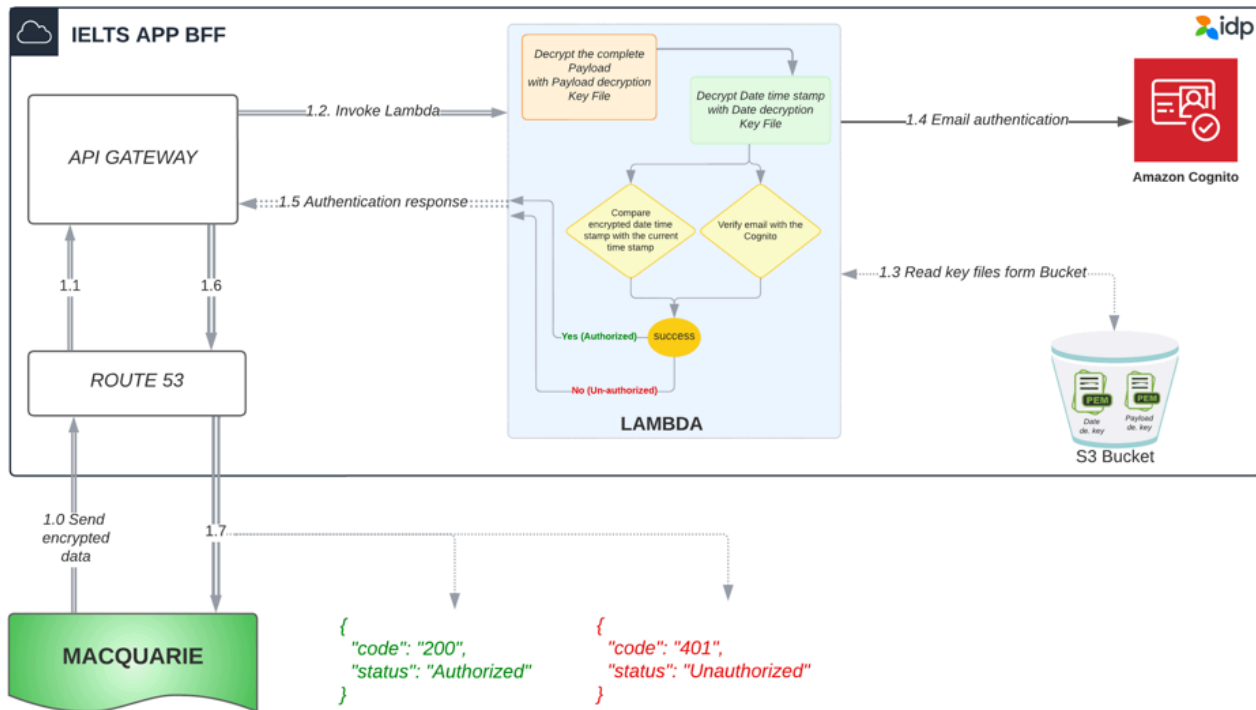
Sample Encrypted Payload

```

1 "CXDXGcrudaooa79A9yE3M0jcDJYeBKoNwR/tRW2xmDOPN9H4mYT60wx25jN0rmEPUdris1HPM6uSQzYWcAsJ9X98wVx+rMd/SRvhW07m1MHzwB+X1/Uyo5G1qvseFefEcVXZS1fvth5zwo50TpAnbNZ2wwhXIXWmknmjSswx1w/Vxo0/jPTgTqeF7BU/VVVex7K27WV0TKDUGpM5Y19YZ+I7SXYFe6My0Er1b2ZUc1i3bowKLfCzZ0ZDqv81fL+LP3r4J6RXjicqAJWHuEiexnRFU2peqe3Gzk45G02wEC6hqi30y/06rGN26CYKqwCDwTf7zN4i7rWtHW609eflg=="
  
```

#3. Macquarie, will receive the encrypted Payload data from in-app browser.

1. Macquarie will do payload cross verification. Macquarie will pass the same payload back to IELTS APP BFF via REST-API call. (POST Request)[[\[35.213.219.236\]](#) IP will be added in the Whitelist]



IELTS APP BFF, API Gateway will receive the encrypted payload and handover to lambda authorize, Lambda authorizer do the below action

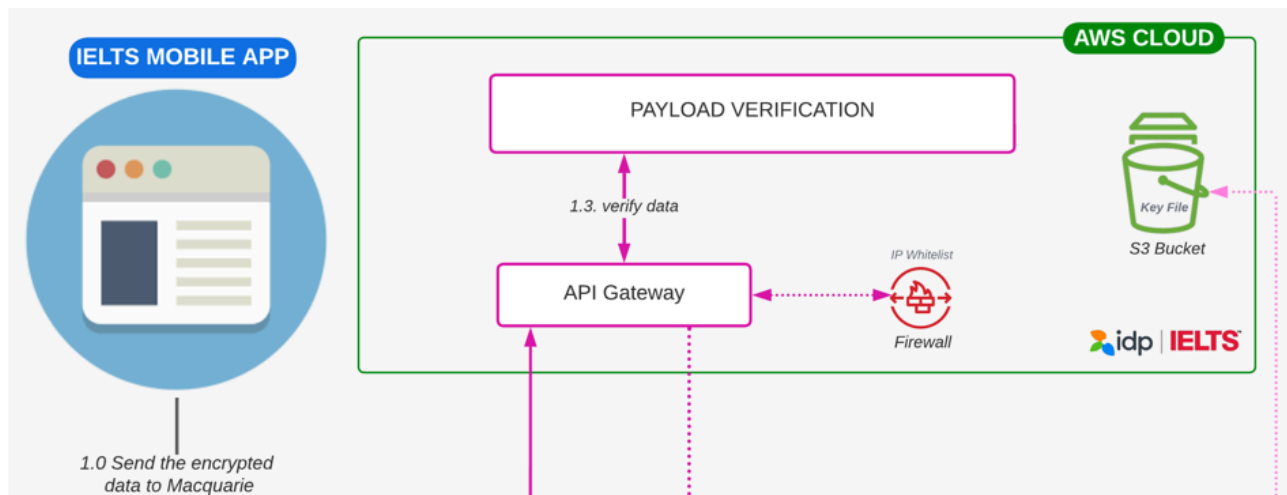
- 1. If the encrypted payload not received via headers, then set **"Effect": "Deny"** else set **"Effect": "Allow"** in the policy response.
- 2. Decrypt the payload with the Payload private .pem file. and get the Payload raw data.
- 3. Decrypt the date time defined in the Raw Payload data.
- 4. Match the email_address with the cognito, if not matched then set **"Effect": "Deny"**, else set **"Effect": "Allow"** in the policy response
- 5. Compare the decrypted date time value with the current date time value, if the time difference is more than 1 hour , if then set **"Effect": "Deny"**, else set **"Effect": "Allow"** in the policy response
- 6. the Policy effect is **"Effect": "Deny"** the return back as **Unauthorized 401** to the Client, if the Policy effect is **"Effect": "Allow"** then return back as **Authorized 200** to the client.

3. Macquarie will received any one of the response from the IELTS APP BFF,

1. if the response status =200

- a. Check the email_address in their database, if exists then auto-login the user within Macquarie
 - b. if the email_address not exists then Step-4
2. If the response status=401
- a. Refresh the in-app browser (to get the new session_id) and repeat the process from **#3**
- 4.** Show the registration page by prepopulating the user details. On Submission, the user will be created in the Macquarie database and auto login.

Decryption logic inside Macquarie



Code Sample:

You can refer the sample code from the below document under - **option-2**

[Authentication: Private Public Key](#)

Certificates:

```

1 Certificate
2   → macquarie.ielts.dev.idp.com
3   → macquarie.ielts.stg.idp.com
4   → macquarie.ielts-stg.idp.com
5   → macquarie.ielts.idp.com
6
7 Domain
8   → https://macquarie.ielts.dev.idp.com
9   → https://macquarie.ielts.stg.idp.com
10  → https://macquarie.ielts-stg.idp.com
11  → https://macquarie.ielts.idp.com

```

References:

Website Link: [Macquarie Group Limited | Global Financial Services](#)

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